Using scrum to Develop a Registry System

This document describes the detailed Scrum software engineering activities, and corresponding example(s), from the beginning of the project until we deliver the complete system to the customer.

1. **Project Initiation and Planning**:
   - **Objective**: Understand project requirements, scope, and constraints.
   - **Activities**:
     - **Stakeholder Analysis**: Identify key stakeholders (users, customers, sponsors).
     - **Vision and Scope Definition**: Define the purpose, features, and boundaries of the gift registry system.
     - **Example**: Imagine stakeholders express the need for features like user registration, item listing, and gift sharing.
2. **Product Backlog Refinement**:
   - **Objective**: Continuously refine and prioritize the Product Backlog.
   - **Activities**:
     - **User Story Refinement**: Break down user stories into smaller tasks.
     - **Estimation**: Estimate effort (story points) for each backlog item.
     - **Example**:
       - **User Story**: “As a user, I want to search for gift items by category.”
       - **Acceptance Criteria**: The search feature should filter items based on selected categories.
3. **Architecture and Design**:
   - **Objective**: Define the system’s architecture and create detailed designs.
   - **Activities**:
     - **High-Level Architecture**:
       - **Example**: Choose a microservices architecture with separate services for user management, item catalog, and registry management.
     - **UI/UX Design**:
       - **Example**: Create wireframes for the registration page, item listing, and gift sharing interfaces.

4. **Sprint Planning**:
    - **Objective**: Plan work for the upcoming sprint (timeboxed iteration).
    - **Activities**:
        - **Select Backlog Items**:
            - **Example**: In Sprint Planning, the team selects user stories like “User registration” and “Add items to registry.”
        - **Task Breakdown**:
            - **Example**: Break down “User registration” into tasks: design UI, implement backend logic, and write tests.
5. **Development (Coding)**:
    - **Objective**: Implement features according to the design.
    - **Activities**:
        - **Pair Programming**:
            - **Example**: Two developers collaborate to write code for the “Add items to registry” feature.
        - **Version Control (Git)**:
            - **Example**: Create a Git branch for the “User authentication” task.
6. **Testing**:
    - **Objective**: Ensure the system works correctly.
    - **Activities**:
        - **Unit Testing**:
            - **Example**: Write unit tests for the “Search items” functionality.
        - **Integration Testing**:
            - **Example**: Verify that adding an item to the registry updates the database correctly.
7. **Sprint Review and Retrospective**:
    - **Objective**: Demonstrate completed work and reflect on the sprint.
    - **Activities**:
        - **Demo**:
            - **Example**: Show the working “User registration” feature to stakeholders.
        - **Retrospective**:
            - **Example**: Discuss what went well (e.g., efficient collaboration) and areas for improvement (e.g., better task estimation).

8. **Release and Deployment**:
    - **Objective**: Prepare for deployment.
    - **Activities**:
        - **Integration Testing**:
            - **Example**: Validate the entire system before deploying to production.
        - **Documentation**:
            - **Example**: Create deployment guides for server setup and database configuration.
9. **Customer Acceptance and Handover**:
    - **Objective**: Obtain customer approval and transition ownership.
    - **Activities**:
        - **User Training**:
            - **Example**: Train end-users on how to create and manage their gift registries.
        - **Handover Documentation**:
            - **Example**: Provide a user manual explaining registry features.

Remember, these examples are illustrative, and your actual gift registry project may have unique requirements. Adapt the training to fit your team's context!